

BORA paper-data: figure/table review

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Table 1: Evaluation datasets: MAL (malware scanning), DNA (genomic variants), and DLP (e-mail DLP). The *Rule shape* column summarizes the form and variety of the regexes.

	Document corpus				Regex set			
	Corpus	Docs	Size	Per-doc (min/med/max)	Ruleset (ver.)	Rules	Size	Rule shape
MAL	CentOS 7	1,209	765 MB	66 KB / 130 KB / 33 MB	ClamAV 0.103.11	38,875	11.6 MB	literal rich; 160,854 leaf subsigs (561 PCRE); logical AND/OR combinations; frequent bounded gaps $\{n\}$; mostly unanchored, implicit $.*\dots*$ shape
DNA	chr17 (GRCh38)	1	83 MB	— (single)	NCBI ClinVar (chr17)	27,500	1.4 MB	anchored $\wedge.\{N\}\text{literal}.*$; len 44–8,738 chars (avg 51)
DLP	Enron Email	509,610	1.38 GB	398 B / 1.5 KB / 1.7 MB	MS-DLP (SIT)	136	10 KB	keyword-proximity: $(kw_1 \dots kw_n)\{0,300\}re re.\{0,300\}(kw_1 \dots kw_n)$

Dataset	Pairs	CP	SDE	DFA
MAL	4.70×10^7	99.8803%	0.1144%	0.0053%
DNA	2.75×10^4	99.9382%	0.0618%	0.0000%
DLP	4.98×10^9	99.9980%	0.0020%	0.0000%

Table 2: Tier-discharge fractions per dataset.

Size (B)	Files	CP	SDE	DFA
$2^{16}\text{--}2^{19}$	954	99.9140%	0.0808%	0.0052%
$2^{19}\text{--}2^{21}$	220	99.7566%	0.2380%	0.0054%
$2^{21}\text{--}2^{23}$	21	99.8119%	0.1809%	0.0072%
$2^{23}\text{--}2^{25}$	14	99.6333%	0.3577%	0.0090%

Table 3: Tier-discharge fractions by file size (Mal).

Size (B)	Files	CP	SDE	DFA
$2^7\text{--}2^{11}$	326235	99.9992%	0.0008%	0.0000%
$2^{11}\text{--}2^{14}$	172337	99.9965%	0.0035%	0.0000%
$2^{14}\text{--}2^{17}$	6130	99.9793%	0.0207%	0.0000%
$2^{17}\text{--}2^{20}$	152	99.9384%	0.0616%	0.0000%

Table 4: Same, over Dlp.

Component	Purpose	MAL		DNA		DLP	
		M	%	M	%	M	%
range table [†]	range proof e.g. for pattern distance	67.1	85.9	134.2	98.4	33.6	99.6
AC-DFA / CP	transition rules	3.2	4.2	1.1	0.8	0.1	0.2
SDE	subsig step pattern distance constraints	7.7	9.9	1.1	0.8	0.1	0.2
DFA (4/0/0)	transition rules	0.1	0.1	0.0	0.0	0.0	0.0
sig-DB	DNF combinations / tri-logic truth table	0.0	0.0	0.0	0.0	0.0	0.0
TOTAL populated		78.2	29.1	136.4	50.8	33.7	12.6
#signatures		1,944		1,375		494	

Table 5: Lookup-table composition across datasets. Column M is millions of entries. Each % is of that dataset’s populated entries; TOTAL % is of the 2^{28} capacity. [†]range table = $2^{\text{range}2.\text{bit}}$ (26 / 27 / 25 for MAL/DNA/DLP), sized by the maximum document offset. The $(\cdot / \cdot / \cdot)$ after *DFA* is the number of DFAs folded in for rules that frequently fail SDE, per dataset. BORA re-writes $(\text{kw1} | \dots | \text{kwN}) \cdot \{0, N\} \text{pat}$ in DLP to disjunction of n rules for improved SDE accuracy, thus expanding from 136 to 9861.

Dataset	Circ	input		R1CS / input byte: raw (adjusted)				corpus
		(B)	R1CS	c_1 (CP)	c_2 (SDE)	c_3 (DFA)	c_4 (frwk)	
MAL	0	3,968	380,454	9.3 (29.0)	18.6 (57.1)	4.5 (4.8)	63.5 (5.0)	3.6%
	1	3,968	435,331	10.4 (27.9)	27.0 (72.0)	4.5 (4.7)	67.9 (5.1)	96.4%
DNA	0	7,936	885,871	28.7 (63.9)	22.6 (45.4)	0.0 (0.0)	60.4 (2.4)	100.0%
DLP	0	1,984	80,140	2.8 (6.8)	9.2 (22.6)	0.0 (0.0)	28.4 (11.0)	49.2%
	1	1,984	103,675	2.9 (5.6)	17.8 (35.6)	0.0 (0.0)	31.6 (11.0)	50.8%

Table 6: Per-circuit cost profile. Each dataset is discharged by a small set of circuits. *input* is the bytes processed per folding step. Each $c_i \in (c_1, c_2, c_3, c_4)$ is the per-input-byte R1CS constraints of the four components of a circuit. Each c_i is shown as *raw (adjusted)*: the *adjusted* value moves the *measured* log-up query cost in c_4 and charges it to each tier in proportion to its data-column size, so $c_1/c_2/c_3$ reflect the all-in per-byte cost and c_4 retains only folding overhead and the processing of Logup share of lookup table.

Dataset	Corpus	Jobs	Steps	Stage 1: Main Folding			Total / Wall (jobs)	Stage 2
				Sel.	Commit wit.	Fold		Compression
MAL	829.2 KB	2	214	47.5 s	1.2 min	4.6 min	6.7 min / 3.3 min (2)	50.5 s
DNA	813.8 KB	1	105	20.3 s	34.7 s	3.4 min	4.3 min (1)	56.3 s
DLP	350.7 KB	2	181	16.2 s	11.9 s	1.6 min	2.1 min / 1.1 min (2)	31.1 s
Per batch proof (one per job, constant): size 1,152 B, verification 24 ms.								

Table 7: End-to-end prover performance. *Corpus* is the bytes actually folded: DNA packs each base of its chr17 sequence into a nibble (half a byte), halving it to the 813.8 KB shown, while the 829.2 KB MAL results from padding each chunk to 128 KB. *Stage 1* (main folding) is the corpus-scaling cost: circuit selection + commit fixed witness + folding. *Total / Wall (jobs)* reports the summed CPU-time over the parallel jobs and the wall-clock time; for balanced jobs the wall-clock is $\approx$ Total / Jobs. *Stage 2: Compression* computes the main-circuit SNARK proof, the QA-NIZK proof, the cycle-pairing fold, and the resulting batch proof. It is a fixed per-job cost, and all timings exclude the one-time key setup. The bottom row shows the size and verification cost of one batch proof (mainly the two Groth16 SNARK proofs); this per-job batch proof is constant size and independent of corpus and rule-set size. A run emits one batch proof per job, as jobs split a corpus to save wall time.

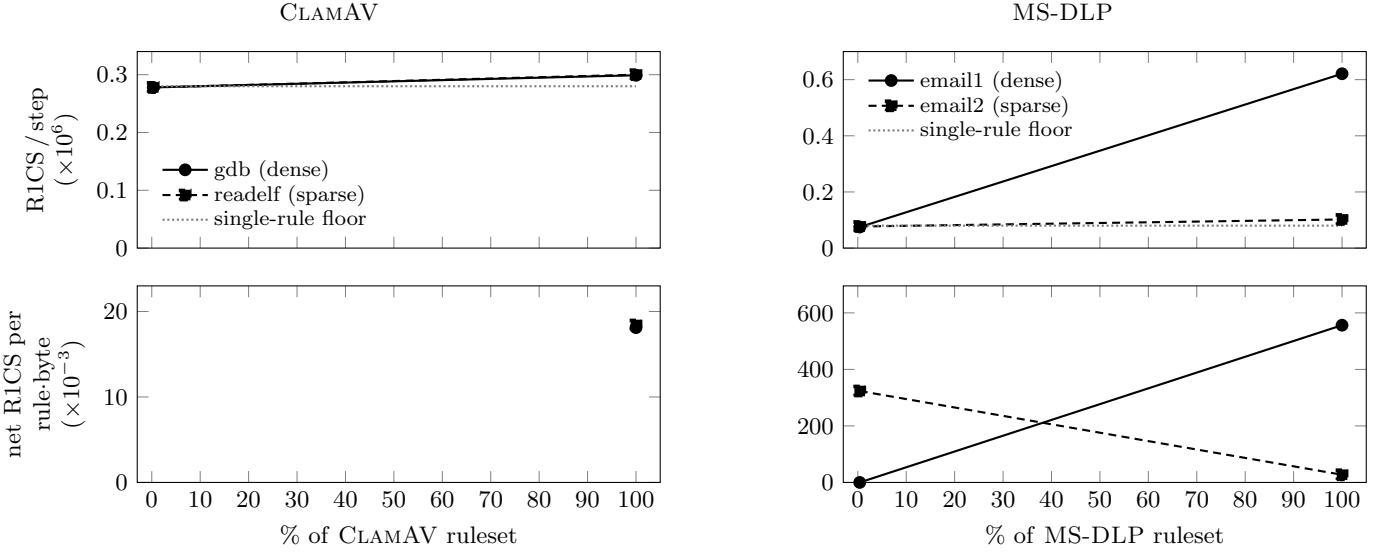


Figure 1: Ruleset-size scalability (input fixed), per folding step, on CLAMAV (left) and MS-DLP (right). *Top*: absolute circuit size with the single-rule framework floor (dotted; 0.28 M R1CS for CLAMAV, 0.08 M for MS-DLP); growing the ruleset to the full 300 (CLAMAV) / 494 (MS-DLP) rules grows the circuit far slower than the rule count. *Bottom*: net circuit cost *per rule per input byte*—cumulative R1CS above the floor divided by cumulative rule count times per-step input length—falling from 18.11×10^{-3} to 18.11×10^{-3} (**gdb**) and 18.42×10^{-3} to 18.42×10^{-3} (**readelf**) R1CS per rule-byte (CLAMAV) and 556.39×10^{-3} to 556.39×10^{-3} (**email1**) and 323.59×10^{-3} to 27.24×10^{-3} (**email2**) R1CS per rule-byte (MS-DLP) at the full set. For the same MS-DLP ruleset, ZOMBIE emits 3,407,872 R1CS in total; dividing by the 2000-byte input and the same 494 rules gives ~ 3.45 R1CS per rule-byte, the same normalization applied to BORA.

Table 8: Reef per-bucket cost vs. BORA on chr17 $\times$ NCBI (27,500 regexes). Both systems are compared on the *net* step-folding cost only, excluding commitment, setup, and IVC proof compression, for a fair comparison. For Reef, net = witness generation + Nova prove + SAFA solve (**fa_solver**); it excludes the fixed floor (the 4.3 GB commitment load, SNARK setup, consistency proof, and related one-time costs). For BORA, net = the Phase 1 circuit selection and non-deterministic advice generation, and Phase 2 main-folding step cost, excluding the Groth16 IVC proof compression as well as setup and commitment. Estimate1 uses each bucket’s measured per-run mean times its population count; Estimate2 uses the minimum measured per-run mean (proj_512k) times bucket count as a counterfactual floor.

Bucket	Count	Per-run net cost (s)	Estimate1 (s)	Estimate2 (s)
non_projectable	24,434	826.98 ± 0.00	2.021×10^7	1.244×10^5
proj_512k	58	5.09 ± 0.00	2.952×10^2	2.952×10^2
proj_2M	132	13.53 ± 0.00	1.786×10^3	6.719×10^2
proj_4M	1,194	25.61 ± 0.00	3.058×10^4	6.077×10^3
proj_8M	869	51.60 ± 0.00	4.484×10^4	4.423×10^3
proj_16M	813	88.43 ± 0.00	7.189×10^4	4.138×10^3
Reef TOTAL	27,500	—	2.036×10^7 (235.60 d)	1.400×10^5 (38.88 hr)
BORA	27,500	— (single batched run)	2.581×10^2 (4.3 min) — $\approx 78869\times$ faster	2.581×10^2 (4.3 min) — $\approx 542.34\times$ faster

Table 9: Zombie prover cost on the MS-DLP international SIT rules (Spartan NIZK, two-pattern proximity circuit), totalled over all 8 regex instances (the 8 SIT rules of Table 1, expanded into their comb variants) at each scanned document length L_T . The total regex size (sum of pattern + keyword bytes over all instances, each sub-pattern counted once) is fixed at 9,600 B. The last row derives the unit cost $u = t_{\text{prove}} / (L_T \cdot \sum s)$ in seconds per (input-byte $\times$ regex-byte). Rows marked † have no real measurement at that length in this log (e.g. a dry-run sweep with a smaller **VEC_SIZE**) – all four totals are linearly interpolated from the 1,000 B row, under a linear-growth assumption matching Zombie’s per-byte circuit complexity; treat them as illustrative, not measured.

L_T (B)	Total constraints	Prove (s)	Verify (s)	Proof (MB)
700	1,572,864	9.4	5.0	0.21
800	1,703,936	11.8	6.4	0.23
1,000	1,703,936	16.9	9.8	0.25
2,000 †	3,407,872	33.8	19.6	0.50
Unit $u = t / (L_T \cdot \sum s)$ at $L_T = 700/800/1,000/2,000$: $1.39 \times 10^{-6} / 1.53 \times 10^{-6} / 1.76 \times 10^{-6} / 1.76 \times 10^{-6}$ s B $^{-2}$				

Table 10: Projected Zombie cost (unit $\bar{u} = 1.76 \times 10^{-6} \text{ sB}^{-2}$, taken at $L_T = 2,000$) vs. BORA’s net main-circuit folding cost. Est. Zombie = $\bar{u} \cdot (\text{corpus}) \cdot (\text{regex set})$, a full document $\times$ regex cross product. Regex-set size is on-disk rule-file bytes for MAL/DNA; for DLP it is the pattern+keyword bytes fed to the circuit (the on-disk `.regex` doubles each sub-pattern for the bidirectional proximity form). $\bar{u}$ itself is linearly interpolated (see Table 9’s $\dagger$ note): this log has no real measurement at $L_T = 2,000$, e.g. a dry-run sweep. The MAL (ClamAV) BORA cost 0.11 hr is the total over its 2 parallel jobs; the wall-clock time is 0.09 hr (the slowest job). The DLP (Enron) BORA cost 0.04 hr is the total over its 2 parallel jobs; the wall-clock time is 0.02 hr (the slowest job).

Dataset	Corpus	Regex set	Est. Zombie	BORA (net)	Speedup
MAL (ClamAV)	765.5 MB	11.6 MB	1.57×10^{10} (497.7 yr)	4.00×10^2 (0.11 hr)	$\approx 39,250,551 \times$
DNA (chr17)	83.3 MB	1.4 MB	2.07×10^8 (6.6 yr)	2.58×10^2 (0.07 hr)	$\approx 803,435 \times$
DLP (Enron)	1.42 GB	9.6 KB	2.40×10^7 (278.2 d)	1.26×10^2 (0.04 hr)	$\approx 190,658 \times$

Table 11: End-to-end prover cost of Zombie [?], Reef [?], and BORA on the three datasets of Table 1; the last two columns give BORA’s speedup over each. Zombie costs are projected from the unit cost $u = 1.76 \times 10^{-6} \text{ sB}^{-2}$ (Table 9). For DNA, Reef’s as-implemented cost is shown with its optimistic floor in parentheses. Values marked * are projected from Reef’s cheapest measured per-run cost on DNA (5.09 s, the `proj_512k` bucket): the DNA floor as that cost per signature, and Reef MAL/DLP as that cost per signature per corpus byte (Table 8). MAL BORA is the total net cost summed over 2 parallel jobs. DLP BORA is the total net cost summed over 2 parallel jobs. Cells shown as “?” have no audited measurement yet.

Dataset	Zombie	Reef	BORA	Speedup	
				vs Zombie	vs Reef
MAL	497.7 yr	4.8 Myr*	0.11 hr	39M $\times$	378B $\times$ *
DNA	6.6 yr	235.6 d (38.88 hr*)	0.07 hr	803,435 $\times$	78,869 $\times$ (542 $\times$ *)
DLP	278.2 d	31,175 yr*	0.04 hr	190,658 $\times$	8B $\times$ *